Bulacan State University
College of Science
MAT 403 - Numerical Analysis

Battery Life Prediction for Mobile Devices Using Numerical Approximation

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By

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Abstract

Battery life is one of the most critical factors affecting the usability and performance of mobile devices. Users rely on smartphones, tablets, and laptops for academic, professional, and personal activities, yet battery depletion often occurs unpredictably due to varying usage patterns and hardware conditions. This study focuses on predicting battery life using numerical approximation techniques based on data collected from students across the campus through Google Forms. The collected data include usage habits, charging behavior, screen time, and application usage. Numerical methods such as finite difference approximation and regression-based numerical modeling are applied to estimate battery discharge rates over time. These methods allow the modeling of battery percentage as a continuous function, even when data are collected at discrete time intervals. The results demonstrate that numerical approximation can provide reasonable predictions of remaining battery life under typical usage scenarios. Such predictions have practical applications in battery management systems, energy-efficient app design, and user decision-making. Overall, this study highlights the relevance of numerical methods in solving real-world engineering and technology problems and shows how mathematical modeling can enhance everyday mobile device usage.

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1 Introduction

Mobile devices have become an important part of everyday life, especially for communication, schoolwork, navigation, and productivity. As people rely more on smartphones, battery performance has become a major concern since it directly affects how long a device can be used. Being able to accurately estimate the remaining battery life helps users manage their time and device usage better. However, many existing battery life prediction methods are still inconsistent and sometimes unreliable (Li et al., 2018). This shows the need for better approaches to predicting battery life in real-world situations.

Battery discharge is affected by several factors that connects with each other, such as how apps are used, level of screen brightness, background activities, and even environmental conditions like temperature. Because of these factors, battery behavior changes over time in nonlinear ways, making it difficult to model using simple formulas. Researchers have explored both physical and data-driven models to better represent battery discharge patterns (Wang et al., 2023). Despite this, traditional analytical models often struggle to handle irregular data and varying usage patterns, which highlights the need for more flexible computational methods.

Numerical analysis has useful mathematical techniques for estimating values based on discrete data, especially when exact solutions are not possible. Methods such as interpolation, regression, and curve fitting make it possible to create approximate models using collected data (Burden et al., 2016). These techniques are commonly used in scientific studies to study and analyze real-world data where exact mathematical models are difficult to apply.

This paper focuses on using numerical approximation techniques to model and predict battery life in mobile devices of students from Bulacan State University. It discusses the mathematical concepts behind the selected numerical methods, applies them to sample battery discharge data, and evaluates how effective they are in estimating remaining battery life. The study highlights the importance of numerical approximation in real-world applications and examines its potential to improve the accuracy of battery life predictions.

2 Preliminaries

This section outlines the key concepts that form the foundation of the study. It begins with the fundamentals of numerical approximation, including functions, discrete data points, and the concept of approximating continuous behavior from sampled values. It then introduces finite difference methods and numerical regression techniques as tools for estimating battery discharge rates over time. Finally, it discusses basic modeling concepts used to analyze battery usage data, which serve as the basis for predicting mobile device battery life under varying usage conditions.

2.1 State of Charge (SoC)

The State of Charge (SoC) represents the available capacity of a battery expressed as a percentage of its maximum capacity. It functions as the "fuel gauge" for energy storage devices and is the main state variable in battery management systems (Piller et al., 2001).

Let $S(t)$ denote the State of Charge at a continuous time t. Then:

$$S(t) \in [0, 1] \quad \text{or} \quad [0\%, 100\%]$$

where:

- $S(t) = 1.0$ (or 100%) corresponds to a fully charged battery.
- $S(t) = 0.0$ (or 0%) corresponds to a completely discharged battery.

2.1.1 Discrete–Time Model

For digital simulation and prediction, a discrete-time model is employed, in which time is sampled at regular intervals of duration Δt (e.g., every minute or hour):

$$S[k] = S(k \cdot \Delta t)$$

Where $k = 0, 1, 2, \ldots$ is the discrete time index.

For example, if $\Delta t = 0.1$ hours (6 minutes), then $S[10] = S(1.0 \text{ hour})$ represents the battery's charge level after exactly one hour. Discrete sampling aligns with computer simulations and reflects how mobile devices periodically report battery percentage.

2.1.2 Prediction Framework

The SoC is linked to user-specific parameters derived from survey data, such as device age and usage patterns. The prediction problem can be stated as:

Given:

- An initial SoC, $S[0]$ (typically 1.0 for 100%).
- A set of personalized parameters p derived from the user's survey responses (device age, usage patterns, etc.).

Find: The discrete time k_{end} such that $S[k_{end}] \leq 0$. The predicted battery life is then $k_{end} \cdot \Delta t$ hours.

The evolution of $S[k]$ is governed by the discharge rate, which depends on the current SoC and user parameters p:

$$S[k+1] = S[k] - \Delta t \cdot D(S[k], p)$$

Where $D(\cdot)$ represents the discharge rate in % per hour. Building this function from survey data using numerical methods is the focus of the following sections.

2.2 Finite Differences and Euler's Method

To approximate the continuous change in SoC, numerical methods are employed. The derivative of the SoC with respect to time is approximated using a forward difference scheme:

$$\frac{dS}{dt} \approx \frac{S[k+1] - S[k]}{\Delta t}$$

Substituting this approximation into the governing differential equation results in the update formula used in Euler's Method:

$$B_{n+1} = B_n - (r \times h)$$

where B_n represents the current battery level and h is the step size. Euler's Method provides a simple yet effective way to estimate battery levels at successive time intervals.

2.3 Multiple Linear Regression and Interaction Terms

This study utilizes Multiple Linear Regression to estimate the discharge rate r based on various influencing factors such as device age, battery capacity, usage intensity, and charging behavior. By incorporating multiple variables, the model becomes more representative of real-world battery usage.

To further improve prediction accuracy, an interaction term known as the *Battery Wear Factor* is introduced. This term captures the combined effect of device age (x_i) and charging frequency (x_j). The regression model is expressed as:

$$T = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n (x_i \cdot x_j) + \varepsilon$$

where $\beta_0, \beta_1, \ldots, \beta_n$ are the regression coefficients and ε represents the error term.

2.4 Linear Interpolation

Linear interpolation is used to estimate unknown battery levels between two known data points. Given two points (t_1, B_1) and (t_2, B_2), the battery level at any intermediate time t is approximated by:

$$B(t) = B_1 + \frac{B_2 - B_1}{t_2 - t_1} \cdot (t - t_1)$$

This method assumes that the change in battery level between the two points is linear, providing a reasonable approximation for short time intervals.

3 Methods and Applications

Based on observed battery discharge data, this study estimates battery life in mobile devices using numerical approximation techniques. Battery levels at intervals between recorded measurements are estimated using linear interpolation, the primary numerical technique used. In this study, battery level data is analyzed as a function of time, with measurements taken at predetermined intervals. An approximation model that depicts how the battery level drops over time is constructed using these recorded information. It is possible to calculate intermediate battery levels and predicted battery duration by using linear interpolation between successive data points. The battery percentage readings that were gathered at various times while the device was being used make up the dataset. These data points form the basis for building the numerical model and producing battery life predictions.

3.1 Model Construction

To estimate the remaining battery life of a mobile device, a mathematical model must be developed to describe how the battery level changes over time. The battery percentage is expressed through the concept of *State of Charge (SoC)*, which represents the remaining battery capacity relative to its maximum capacity.

Let $B(t)$ denote the battery level of the device at time t, where $B(t)$ is expressed as a percentage ranging from 0% to 100%. As the device is used, energy is consumed by hardware components such as the screen, processor, and cellular radios. This consumption causes the SoC to decrease gradually.

The model makes the assumption that the device's average power consumption determines the rate of decrease in order to depict this behavior. Let r be the battery's discharge rate, expressed as the percentage of battery loss per unit of time. A first-order ordinary differential equation is used to simulate the change in battery level under this assumption:

$$\frac{dB}{dt} = -r$$

The battery level in this equation drops with increasing time, as indicated by the negative sign. This simplified formulation is an essential beginning point for numerical approximation, even if real battery behavior is non-linear and varies depending on instantaneous tasks.

In a hybrid framework, r is a function of user-specific characteristics rather than a single constant for all devices. These factors include usage intensity, battery capacity, and device age. r can be defined mathematically as:

$$r = f(p_1, p_2, \ldots, p_n)$$

where p stands for the factors, like the Battery Wear Factor, that were gathered from survey data. The model transitions from a generic analytical formula to a customized numerical prediction by using regression analysis to determine r.

The continuous model must be transformed into a discrete form since it is challenging to find precise analytical solutions when the discharge rate varies under various usage scenarios. If battery levels are measured at intervals of size h, the derivative can be estimated using a finite difference approach as follows:

$$\frac{dB}{dt} \approx \frac{B_{n+1} - B_n}{h}$$

Substituting this approximation into the original differential equation leads to the numerical update formula:

$$B_{n+1} = B_n - r \cdot h$$

Euler's Method is based on this discrete representation. By deducting the anticipated energy consumption over each interval h, it enables the program to mimic battery drain step-by-step.

This methodology uses linear interpolation to approximate battery levels at particular points between the discrete time steps and linear regression to calibrate the discharge rate r in addition to predicting battery levels across time. By combining these methods, a useful model is created that makes use of computed discharge rates, present battery condition, and usage trends to predict future performance.

3.2 Hybrid Numerical Prediction Algorithm

Combining multiple numerical methods that can simulate depletion of batteries and take user behavior changes into consideration is necessary to predict a mobile device's remaining battery life. This study

integrates Euler's Method, linear regression, linear interpolation, and finite difference approximations to create a hybrid framework. Every technique advances a distinct phase of the forecasting process.

The battery level as a function of time is the model's starting point. Let the battery

$$\frac{dB}{dt} = -r,$$

where r represents the battery discharge rate.

The derivative must be mathematically calculated because battery data are usually observed at discrete time periods rather than constantly. The pace at which the battery level changes can be expressed using the finite difference approximation as

$$\frac{dB}{dt} \approx \frac{B_{n+1} - B_n}{h},$$

where h is the time interval between measurements and B_n is the battery level at time step n. This approach makes it possible to predict the discharge rate given battery data that has been observed.

The future battery level can be simulated using Euler's Method once the discharge rate has been estimated. The numerical update formula is obtained by replacing the differential equation with the finite difference approximation.

$$B_{n+1} = B_n - rh.$$

By deducting the expected energy usage across the period h, this equation forecasts the battery

In reality, the discharge rate r can change based on user activity and device features. The association between battery lifetime and pertinent usage variables derived from survey data is estimated using linear regression to take these aspects into consideration. The rate of battery consumption can be influenced by factors like battery capacity, daily usage hours, and device age. The regression model may be written as

$$T = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon,$$

where $x_1, x_2, \ldots, x_n$ are variables related to consumption and T is the anticipated battery lifetime. An estimate of the anticipated battery lifetime is provided by the regression model that results, and this estimate can be used to calculate an approximate discharge rate.

When direct measurements are unavailable, it is occasionally required to estimate battery values in addition to determining discharge rates and forecasting future battery levels. For this, linear interpolation is employed. The battery level at an intermediate time t can be predicted if the battery level is known at two time points (t_0, B_0) and (t_1, B_1).

$$B(t) = B_0 + \frac{B_1 - B_0}{t_1 - t_0}(t - t_0).$$

This interpolation offers a straightforward way to maintain continuity in the anticipated battery discharge curve while estimating missing data points.

The hybrid framework offers a methodical approach to battery life prediction by combining different numerical methods. Regression analysis calculates the impact of usage variables on battery lifetime, interpolation closes gaps in observed data, Euler's Method produces a step-by-step forecast of battery levels over time, and finite differences approximate the rate of battery discharge. When combined, these techniques provide a unified numerical approach for assessing and predicting mobile device battery performance.

3.3 Example Simulations

A MATLAB computer simulation was created to show how the suggested hybrid numerical technique could be used in real-world scenarios. Because of its strong capabilities for statistical modeling, matrix operations, and iterative numerical approximations, MATLAB was chosen. By processing the discrete survey data and producing a continuous, predictive battery discharge curve for every user, the simulation implements the mathematical architecture outlined in Section 3.2.

The simulation process is divided into four primary computational phases:

1. **Data Initialization and Wear Factor Formulation.** The survey data arrays are first consumed by the program. A derived computational variable called the Battery Wear Factor is introduced since physical charging practices have a significant impact on battery degradation. This is computed by multiplying the user's daily charging frequency by the age category of the device. Without the need for precise chemical measurements, the numerical model may take into account the physical depreciation of the lithium-ion cells by include this interaction term.

2. **Robust Regression Phase.** The simulation must first estimate the whole projected battery lifetime (T) for a particular user in order to get the baseline discharge rate. The approach applies robust fitting choices using the fitlm function in MATLAB. In order to ensure that the regression model accurately depicts the true physical average, this numerical weighting technique is essential for survey-based data since it algorithmically detects and downgrades the impact of extreme outliers or mistaken human evaluations.

3. **Execution of Euler's Method.** The constant discharge rate (r) is defined as 100 / T once the estimated total lifetime has been determined. Euler's Method is then used to discretize the continuous differential equation modeling battery drain. The simulation employs a discrete time step of h = 0.5 hours (30 minutes) after initializing the State of Charge at 100%. Until the battery hits 0%, the iterative

update formula is run inside a limited loop, producing an array of anticipated battery percentages every 30 minutes.

4. **Linear Interpolation for Continuous Estimation.** The simulation uses linear interpolation because mobile device users might need to forecast their battery life at periods that do not precisely fall on the 30-minute discrete steps. For instance, the system finds the discrete outputs for 6.0 and 7.0 hours and calculates the intermediate value to estimate the battery percentage at precisely 6.5 hours of usage. This guarantees that the prediction model will continue to be continuous and extremely adaptable for actual user requests. Figure 1 below shows the numerical simulation's visual result for users with different discharge rates.

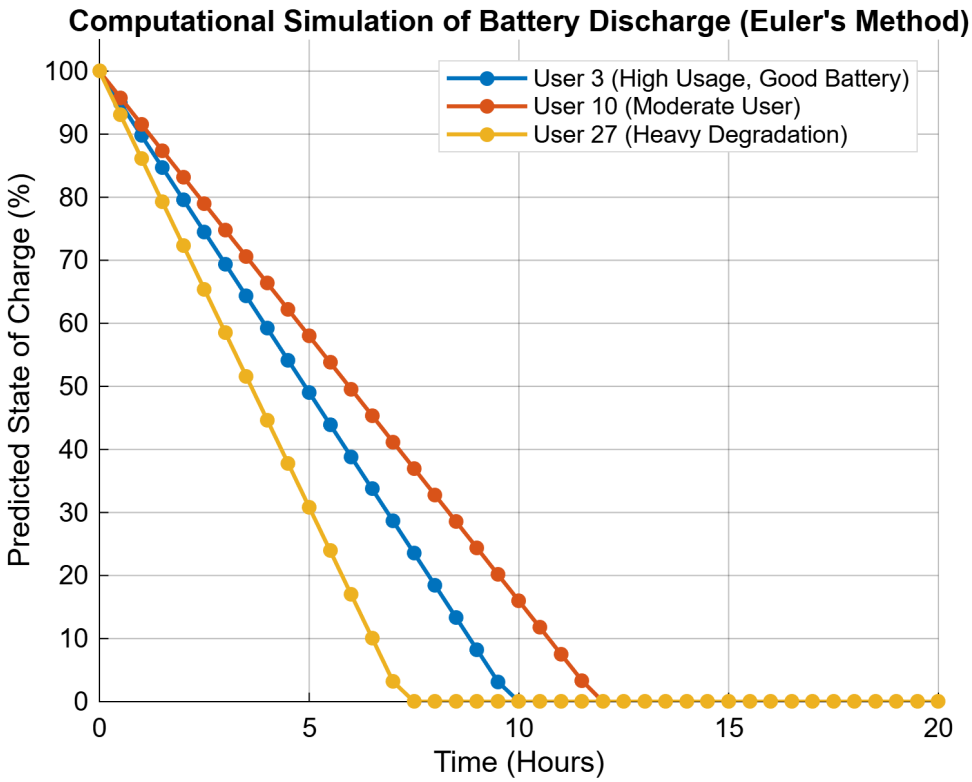

Figure 1: *MATLAB computational simulation of battery discharge using Euler's Method with a step size of $h = 0.5$ hours.*

3.4 Error Analysis & Validation

To validate the accuracy and reliability of the proposed hybrid numerical prediction algorithm, an error analysis was conducted by comparing the model's simulated outputs against the observed real-world data from the survey respondents.

This study used a purely objective physical indicator, the actual end-of-day battery

The Mean Absolute Error (MAE) for each of the 35 users was used to assess the numerical model's overall correctness. An MAE of 10.40% was obtained from the computational simulation. With an average variance of roughly 10 percentage points, this data shows that the numerical simulation was

successful in predicting the user's remaining battery level. Table 1 shows an example of the computational findings compared to real data.

Table 1: Excerpt of Computational Simulation Results vs. Actual Observations

User	Predicted Total Life (Hrs)	Discharge Rate (%/hr)	SoC at 6h (%)	Interpolated SoC at 6.5h (%)	SoC at 7h (%)	SoC at 8h (%)	Euler Pred. at End of Day (%)	Actual End of Day (%)	Error Margin (%)
1	10.67	9.38	43.7	39.1	34.4	25.0	34.4	40	5.63
2	8.34	11.99	28.0	22.0	16.0	4.1	16.0	5	11.05
3	9.81	10.20	38.8	33.7	28.6	18.4	8.2	20	11.78
4	7.76	12.89	22.7	16.2	9.8	0.0	48.5	60	11.55
5	11.02	9.07	45.6	41.0	36.5	27.4	18.3	5	13.35

An average error margin of 10.40% offers a highly accurate numerical validation given the inherent unpredictability of human mobile device usage, such as fluctuating screen brightness, background application consumption, and the constraints of self-reported survey data. The model's success demonstrates how successful the hybrid strategy is. The program created a very realistic baseline discharge rate by combining the physical relationship between device age and charging frequency (Battery Wear Factor) and using strong linear regression to eliminate extreme outliers. Additionally, a flexible, continuous framework that accurately mimicked real-world battery depletion was produced by combining Euler's Method with linear interpolation. In the end, our error analysis demonstrates that numerical approximation methods can greatly enhance the mathematical modeling of mobile device battery life when appropriately bound by empirical usage characteristics.

3.5 Discussion of Limitations

Although the hybrid algorithm achieved a solid 10.40% Mean Absolute Error, there are several notable limitations worth discussing.

The fundamental differential equation ($dB/dt = -r$), which presumes a linear battery discharge, presents a major obstacle. Lithium-ion batteries actually have non-linear discharge. Abrupt, erratic power decreases are caused by things like high temperatures and heavy application usage (like gaming). These quick, minute-to-minute variations are not captured by a constant discharge rate (r), even if it offers a functional average.

Additional restrictions are introduced by the nature of the survey data. Although extreme or implausible replies were filtered using robust regression, human memory and estimation are still used in the study. The baseline data is intrinsically less reliable than data from automatic system logs because consumers frequently have trouble estimating active screen time.

Additionally, the model ignores notable variations in hardware and software by treating all mobile devices as a single group. Power consumption and background processes are managed at varying rates by different chipsets and operating system optimizations. A certain amount of deviation in the projections

is expected since the mathematical model does not account for particular processor brands and software efficiency.

Euler's Method was used for the simulation itself, with a step size of 30 minutes ($h = 0.5$). This method produces a less refined discharge curve while maintaining computational efficiency. A smoother simulation and possibly lower interpolation errors could be achieved by using a smaller time step (e.g., $h = 0.1$) or more sophisticated numerical approaches, like the Runge-Kutta (RK4) method.

The ultimate goal of this expository study, which is centered on numerical analysis, is not to take the place of the complex physical battery trackers present in contemporary gear. Instead, the MATLAB simulation is a useful example of how abstract mathematical ideas, such linear interpolation and finite differences, may be used to estimate complicated real-world variables.

4 Conclusion

The ability to accurately anticipate battery life is extremely useful because smartphones are an integral part of daily life. This study shows that using numerical approximation to empirical usage data provides a workable solution, even though conventional, inflexible formulas may find it difficult to address unexpected human behavior.

Complex survey responses were successfully converted into a working mathematical framework by connecting a strong regression model with Euler's Method and linear interpolation. The notion was validated by testing this hybrid approach against data from 35 university students, as the model predicted end-of-day battery levels with an average error of only 10.40

This level of accuracy demonstrates the adaptability of numerical approaches, especially when using self-reported data. The study developed a useful technique for battery life estimation by taking into consideration physical patterns, such as the relationship between device age and charging frequency, and using iterative simulations. Overall, this study demonstrates that numerical analysis is a potent tool for resolving technical issues in real-world applications rather than just abstract theory.

References

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Li, H., Liu, X., & Mei, Q. (2018). Predicting smartphone battery life based on comprehensive and real-time usage data. *arXiv preprint arXiv:1801.04069*. https://arxiv.org/abs/1801.04069

Piller, S., Perrin, M., & Jossen, A. (2001). Methods for state-of-charge determination and their applications. *Journal of Power Sources*.

Wang, X., Ye, P., Liu, S., Zhu, Y., Deng, Y., & Yuan, Y. (2023). Research progress of battery life prediction methods based on physical model. *Energies*, *16*(9), 3858. https://doi.org/10.3390/en16093858

A MATLAB Source Code

The following script was developed to implement the hybrid numerical prediction model. It performs data initialization, robust linear regression, iterative simulation via Euler's Method, and linear interpolation to predict State of Charge at specific intervals.

```
% Battery Life Prediction for Mobile Devices Using Numerical Approximation
% Hybrid Numerical Prediction Algorithm (Regression + Euler + Interpolation)

clc; clear; close all;

%% --- Step 1: Numeric survey data for 35 users ---
survey_data = [
7, 1, 3582, 2, 5, 0, 40, 6;   7, 3, 5000, 2, 2, 1, 5, 6;    9, 1, 5000, 1, 2, 1, 20, 15;
4, 3, 4500, 2, 5, 0, 60, 10;  9, 3, 5000, 2, 5, 1, 5, 6;    7, 3, 3227, 2, 3, 1, 20, 6;
7, 0, 6000, 1, 5, 1, 20, 15;  7, 3, 2942, 2, 2, 1, 20, 6;   9, 1, 3349, 1, 4, 1, 40, 15;
9, 3, 5000, 2, 4, 0, 20, 6;   9, 3, 3110, 2, 2, 1, 20, 6;   9, 1, 5000, 2, 5, 1, 20, 6;
7, 3, 3110, 2, 1, 1, 20, 6;   7, 0, 3110, 1, 2, 0, 20, 10;  7, 0, 5000, 1, 2, 0, 60, 10;
9, 3, 4352, 2, 3, 1, 5, 10;   7, 1, 3000, 2, 3, 1, 5, 6;    7, 3, 3110, 2, 4, 1, 60, 6;
7, 3, 3227, 2, 2, 1, 5, 6;    7, 1, 5160, 1, 3, 0, 5, 6;    7, 3, 5000, 2, 4, 1, 20, 15;
9, 1, 3279, 2, 2, 1, 20, 6;   9, 3, 3110, 2, 5, 1, 20, 6;   9, 3, 5000, 2, 3, 1, 20, 10;
9, 3, 5000, 2, 5, 1, 20, 10;  7, 3, 2815, 2, 4, 1, 20, 6;   4, 0, 5000, 1, 3, 0, 60, 10;
9, 0, 5160, 1, 2, 0, 20, 10;  9, 3, 2942, 2, 2, 1, 20, 6;   4, 0, 4674, 1, 3, 1, 20, 10;
9, 1, 5000, 2, 4, 0, 20, 15;  9, 3, 3227, 2, 3, 1, 60, 10;  4, 3, 4000, 1, 1, 0, 20, 6;
9, 3, 6000, 2, 2, 1, 20, 6;   7, 1, 4565, 2, 2, 1, 20, 6
];

X_base = survey_data(:,1:6);
num_users = size(X_base,1);

%% --- Step 2: Calculate TRUE Physical Battery Life for Training ---
Usage_Hours = survey_data(:,1);
End_of_Day_Pct = survey_data(:,7);

True_Discharge_Rate = (100 - End_of_Day_Pct) ./ Usage_Hours;
True_Battery_Life = 100 ./ True_Discharge_Rate;
Y_target = True_Battery_Life;

%% --- Step 3: Fit Advanced Regression Model ---
DeviceAge = X_base(:, 2);
ChargeFreq = X_base(:, 4);
BatteryWearFactor = DeviceAge .* ChargeFreq;
X_advanced = [X_base, BatteryWearFactor];

```

```
mdl = fitlm(X_advanced, Y_target, 'RobustOpts', 'on');

%% --- Step 4: Define Time Parameters for Euler's Method ---
h = 0.5;                % Discrete time step (30 minutes)
time_grid = 0:h:24;     % Simulate up to 24 hours
num_steps = length(time_grid);

%% --- Step 5: Setup Results Table ---
BatteryTable = table('Size',[num_users,10], ...
    'VariableTypes', repmat({'double'}, 1, 10), ...
    'VariableNames',{'User','PredTotalLife','DischargeRate_r','Battery6h','Interpolated_6_5h','Battery7h','Battery8h','ActualEndOfDayPct','EulerPredAtEndOfDay','ErrorMargin_Pct'});

%% --- Step 6: Compute predictions using Euler's Method ---
for i = 1:num_users
    user_vars = X_base(i,:);
    actual_usage_hrs = Usage_Hours(i);
    actual_end_pct = End_of_Day_Pct(i);

    user_wear_factor = user_vars(2) * user_vars(4);
    user_vars_advanced = [user_vars, user_wear_factor];

    T_pred = predict(mdl, user_vars_advanced);
    T_pred = max(3, min(T_pred, 48));
    r = 100 / T_pred;

    B = zeros(1, num_steps);
    B(1) = 100;

    for n = 1:(num_steps-1)
        B(n+1) = B(n) - (r * h);
        if B(n+1) < 0, B(n+1) = 0; end
    end

    B_6h = B(time_grid == 6);
    B_7h = B(time_grid == 7);
    B_8h = B(time_grid == 8);

    % Linear Interpolation at 6.5 hours
    B_6_5h = B_6h + ((B_7h - B_6h) / (7 - 6)) * (6.5 - 6);
    if B_6_5h < 0, B_6_5h = 0; end

    Euler_Pred_At_Usage = 100 - (r * actual_usage_hrs);
```

```
    if Euler_Pred_At_Usage < 0, Euler_Pred_At_Usage = 0; end

    BatteryTable.User(i) = i;
    BatteryTable.PredTotalLife(i) = round(T_pred, 2);
    BatteryTable.DischargeRate_r(i) = round(r, 2);
    BatteryTable.Battery6h(i) = round(B_6h, 1);
    BatteryTable.Interpolated_6_5h(i) = round(B_6_5h, 1);
    BatteryTable.Battery7h(i) = round(B_7h, 1);
    BatteryTable.Battery8h(i) = round(B_8h, 1);
    BatteryTable.ActualEndOfDayPct(i) = actual_end_pct;
    BatteryTable.EulerPredAtEndOfDay(i) = round(Euler_Pred_At_Usage, 1);
    BatteryTable.ErrorMargin_Pct(i) = round(abs(Euler_Pred_At_Usage - actual_end_pct),
    2);
end

%% --- Step 7: Display table and Accuracy Metrics ---
disp('--- Battery Life Prediction Results ---');
disp(BatteryTable);

Mean_Error = mean(BatteryTable.ErrorMargin_Pct);
fprintf('\nMean Absolute Error (MAE): %.2f %%\n', Mean_Error);

%% --- EXTRA: Generate a Graph ---
figure('Name', 'Euler''s Method Battery Simulation', 'Color', 'w');
hold on; grid on;

users_to_plot = [3, 10, 27];
colors = {'#0072BD', '#D95319', '#EDB120'};

for k = 1:3
    u_idx = users_to_plot(k);
    T_p = BatteryTable.PredTotalLife(u_idx);
    r_p = 100 / T_p;

    B_plot = zeros(1, num_steps);
    B_plot(1) = 100;
    for n = 1:(num_steps-1)
        B_plot(n+1) = max(0, B_plot(n) - (r_p * h));
    end

    plot(time_grid, B_plot, '-o', 'Color', colors{k}, 'LineWidth', 1.5, ...
        'MarkerSize', 4, 'MarkerFaceColor', colors{k});
end

```

```
title('Computational Simulation of Battery Discharge (Euler''s Method)');
xlabel('Time (Hours)');
ylabel('Predicted State of Charge (%)');
legend('User 3 (High Usage, Good Battery)', 'User 10 (Moderate User)', 'User 27 (Heavy
    Degradation)', 'Location', 'northeast');
xlim([0 20]);
ylim([0 105]);
hold off;

exportgraphics(gcf, 'BatterySimulation.png', 'Resolution', 300);
```

B Full Computational Results Table

The following table contains the complete numerical output for all 35 survey respondents included in the study, showcasing the discrete Euler steps at 6, 7, and 8 hours, the interpolated State of Charge (SoC) at 6.5 hours, and the final prediction error margins.

Table 2: Complete Output of the MATLAB Numerical Simulation

User	Pred. Life (h)	Rate (%/h)	SoC 6h (%)	Interp. 6.5h (%)	SoC 7h (%)	SoC 8h (%)	Actual End (%)	Euler Pred. End (%)	Error Margin (%)
1	10.67	9.38	43.7	39.1	34.4	25.0	40	34.4	5.63
2	8.34	11.99	28.0	22.0	16.0	4.1	5	16.0	11.05
3	9.81	10.20	38.8	33.7	28.6	18.4	20	8.2	11.78
4	7.76	12.89	22.7	16.2	9.8	0.0	60	48.5	11.55
5	11.02	9.07	45.6	41.0	36.5	27.4	5	18.3	13.35
6	8.90	11.24	32.6	27.0	21.3	10.1	20	21.3	1.34
7	9.21	10.86	34.8	29.4	24.0	13.1	20	24.0	3.96
8	8.65	11.56	30.6	24.8	19.1	7.5	20	19.1	0.94
9	10.64	9.40	43.6	38.9	34.2	24.8	40	15.4	24.57
10	11.90	8.40	49.6	45.4	41.2	32.8	20	24.4	4.36
11	10.43	9.59	42.5	37.7	32.9	23.3	20	13.7	6.32
12	11.08	9.02	45.9	41.4	36.9	27.8	20	18.8	1.19
13	8.33	12.01	28.0	22.0	16.0	4.0	20	16.0	4.04
14	9.93	10.07	39.6	34.5	29.5	19.4	20	29.5	9.51
15	9.65	10.37	37.8	32.6	27.4	17.1	60	27.4	32.57
16	10.53	9.49	43.0	38.3	33.5	24.0	5	14.6	9.55
17	9.00	11.12	33.3	27.7	22.2	11.1	5	22.2	17.19
18	9.21	10.86	34.9	29.4	24.0	13.1	60	24.0	36.00
19	8.61	11.62	30.3	24.5	18.7	7.0	5	18.7	13.66
20	9.44	10.59	36.5	31.2	25.9	15.3	5	25.9	20.87
21	8.93	11.20	32.8	27.2	21.6	10.4	20	21.6	1.57
22	10.46	9.56	42.7	37.9	33.1	23.5	20	14.0	6.01
23	11.31	8.84	46.9	42.5	38.1	29.3	20	20.4	0.41
24	10.44	9.58	42.5	37.7	32.9	23.3	20	13.8	6.25
25	11.02	9.07	45.6	41.0	36.5	27.4	20	18.3	1.65
26	9.25	10.80	35.2	29.8	24.4	13.6	20	24.4	4.37
27	7.23	13.82	17.1	10.1	3.2	0.0	60	44.7	15.30
28	11.42	8.75	47.5	43.1	38.7	30.0	20	21.2	1.22
29	10.45	9.57	42.6	37.8	33.0	23.5	20	13.9	6.11
30	6.11	16.36	1.8	0.9	0.0	0.0	20	34.6	14.56
31	11.96	8.36	49.8	45.7	41.5	33.1	20	24.8	4.76
32	10.70	9.34	43.9	39.3	34.6	25.3	60	15.9	44.09
33	5.38	18.59	0.0	0.0	0.0	0.0	20	25.6	5.65
34	9.99	10.01	39.9	34.9	29.9	19.9	20	9.9	10.09
35	8.47	11.81	29.1	23.2	17.3	5.5	20	17.3	2.68